REPUBLIC OF KENYA

MINISTRY OF EDUCATION

JUNIOR SECONDARY SCHOOL CURRICULUM DESIGN

INTEGRATED SCIENCE

GRADE 8

KENYA INSTITUTE OF CURRICULUM DEVELOPMENT

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FOREWORD

The Government of Kenya is committed to ensuring that policy objectives for Education, Training and Research meet the aspirations of the Kenya Constitution 2010, the Kenya Vision 2030, National Curriculum Policy 2019, the United Nations Sustainable Development Goals (SDGs) and the Regional and Global conventions to which Kenya is a signatory. Towards achieving the mission of Basic Education, the Ministry of Education (MoE) has successfully and progressively rolled out the implementation of the Competency Based Curriculum (CBC) at Pre-Primary and Primary School levels. The roll out of Junior Secondary School (Grade 7-9) will subsequently follow as from 2023-2025.

The Grade 8 curriculum designs build on competencies attained by learners at the end of Grade 7. Further, they provide opportunities for learners to continue exploring and nurturing their potentials as they prepare to transit to Senior Secondary School.

The curriculum designs present National Goals of Education, essence statements, general and specific expected learning outcomes for the learning areas (subjects) as well as strands and sub strands. The designs also outline suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values, Community Service Learning (CSL) activities and assessment rubric.

It is my hope that all Government agencies and other stakeholders in Education will use the designs to plan for effective and efficient implementation of the CBC.

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PREFACE

The Ministry of Education (MoE) is implementing the second phase of the curriculum reforms with the national roll out of the Competency Based Curriculum (CBC) having been implemented in 2019. Grade 8 is the second level of the Junior Secondary School (JSS) in the new education structure.

Grade 8 curriculum furthers implementation of the CBC from Grade 7. The main feature of this level is a broad curriculum for the learner to explore talents, interests and abilities before selection of pathways and tracks at the Senior Secondary education level. This is very critical in the realisation of the Vision and Mission of the on-going curriculum reforms as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya. The Sessional Paper explains the shift from a Content - Focused Curriculum to a focus on **Nurturing every Learner's potential.**

Therefore, the Grade 8 curriculum designs are intended to enhance the learners' development in the CBC core competencies, namely: Communication and Collaboration, Critical Thinking and Problem Solving, Creativity and Imagination, Citizenship, Digital Literacy, Learning to Learn and Self-efficacy.

The curriculum designs provide suggestions for interactive and differentiated learning experiences linked to the various sub strands and the other aspects of the CBC. The curriculum designs also offer several suggested learning resources and a variety of assessment techniques. It is expected that the designs will guide teachers to effectively facilitate learners to attain the expected learning outcomes for Grade 8 and prepare them for smooth transition to the next Grade. Furthermore, it is my hope that teachers will use the designs to make learning interesting, exciting and enjoyable.

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ACKNOWLEDGEMENT

The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 (Revised 2019) mandates the Institute to develop curricula and curriculum support materials for basic and tertiary education and training. The curriculum development process for any level of education involves thorough research, international benchmarking and robust stakeholder engagement. Through a systematic and consultative process, the KICD conceptualised the Competency Based Curriculum (CBC) as captured in the *Basic Education Curriculum Framework* (BECF), that responds to the demands of the 21st Century and the aspirations captured in the Kenya Constitution 2010, the Kenya Vision 2030, East African Community Protocol and the United Nations Sustainable Development Goals (SDGs).

KICD receives its funding from the Government of Kenya to enable the successful achievement of the stipulated mandate and implementation of the Government and Sector (Ministry of Education (MoE) plans. The Institute also receives support from development partners targeting specific programmes. The Grade 8 curriculum designs have been developed with the support of the World Bank through the Kenya Secondary Education Quality Improvement Program (SEQIP) commissioned by the MoE. Therefore, the Institute is very grateful for the support of the Government of Kenya, through the MoE and the development partners for the policy, resource and logistical support. Specifically, special thanks to the Cabinet Secretary – MoE and the Principal Secretary – State Department of Early Learning and Basic Education,

We also wish to acknowledge the KICD curriculum developers and other staff, all teachers, educators who took part as panelists; the Semi-Autonomous Government Agencies (SAGAs) and representatives of various stakeholders for their roles in the development of the Grade 8 curriculum designs. In relation to this, we acknowledge the support of the –Chief Executive Officers of the Teachers Service Commission (TSC) and the Kenya National Examinations Council (KNEC) for their support in the process of developing these designs.

Finally, we are very grateful to the KICD Council Chairperson Prof. Elishiba Kimani and other members of the Council for very consistent guidance in the process. We assure all teachers, parents and other stakeholders that these curriculum designs will effectively guide the implementation of the CBC at Grade 8 and preparation of learners for Grade 9.

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LESSON ALLOCATION

	Subject	Number of Lessons Per Week (40 minutes per lesson)
1.	English	5
2.	Kiswahili/KSL	4
3.	Mathematics	5
4.	Integrated Science	5
5.	Pre-Technical Studies	4
6.	Social Studies	4
7.	Religious Education (CRE/IRE/HRE)	3
8.	Business Studies	3
9.	Agriculture	3
10.	Physical Education and Sports	2
11.	Optional Subject	3
12.	Optional Subject	3
	Total	**44**

NATIONAL GOALS OF EDUCATION

Education in Kenya should:

i) **Foster nationalism and patriotism and promote national unity.**

Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by removing conflicts and promoting positive attitudes of mutual respect which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

ii) **Promote the social, economic, technological and industrial needs for national development.**

Education should prepare the youth of the country to play an effective and productive role in the life of the nation.

a) **Social Needs**

Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution following in the wake of rapid modernization. Education should assist our youth to adapt to this change.

b) **Economic Needs**

Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building up a modern and independent economy which is in need of an adequate and relevant domestic workforce.

c) **Technological and Industrial Needs**

Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognizes the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.

iii) **Promote individual development and self-fulfilment**

Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character.

iv) **Promote sound moral and religious values.**
Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant and integrated citizens.

v) **Promote social equality and responsibility.**
Education should promote social equality and foster a sense of social responsibility within an education system which provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.

vi) **Promote respect for and development of Kenya's rich and varied cultures.**
Education should instill in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development in order to build a stable and modern society.

vii) **Promote international consciousness and foster positive attitudes towards other nations.**
Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights and benefits that this membership entails.

viii) **Promote positive attitudes towards good health and environmental protection.**
Education should inculcate in young people the value of good health in order for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

LEARNING OUTCOMES FOR MIDDLE SCHOOL

By end of Middle School, the learner should be able to:

1. Apply literacy, numeracy and logical thinking skills for appropriate self-expression.
2. Communicate effectively, verbally and non-verbally, in diverse contexts.
3. Demonstrate social skills, and spiritual and moral values for peaceful co-existence.
4. Explore, manipulate, manage and conserve the environment effectively for learning and sustainable development.
5. Practise relevant hygiene, sanitation and nutrition skills to promote health.
6. Demonstrate ethical behaviour and exhibit good citizenship as a civic responsibility.
7. Appreciate the country's rich and diverse cultural heritage for harmonious co-existence.
8. Manage pertinent and contemporary issues in society effectively.
9. Apply digital literacy skills for communication and learning.

ESSENCE STATEMENT

Integrated Science is a new learning area that enable learners to apply distinctive ways of logical valuing, thinking and working to understand natural phenomena in the biological, physical and technological world. The learning area is expected to create a scientific culture that inculcates scientific literacy to enable learners to make informed choices in their personal lives and approach life challenges in a systematic and logical manner. The inclusion of Integrated Science is therefore a deliberate effort to enhance the level of scientific literacy of all learners and equip them with the relevant basic integrated scientific knowledge, skills, values and attitudes needed for their own survival and/or career development. Concepts in Integrated Science are presented as units within which there are specific strands that build on the competencies acquired in Science and Technology at Upper Primary level. The emphasis of science education at lower secondary levels is to enhance learners' scientific thinking through learning activities that involve the basic science process skills.

Integrated Science provides the learner with the basic requisite skills, knowledge, values and attitudes necessary for specialization in STEM pathway at senior school level. The rationale for inclusion of Integrated Science is anchored on the

Kenya Vision 2030, Sessional Papers No. 14 of 2012, and No. 1 of 2019, which all underscore the importance of science, technology and innovation in education and training.

Integrated Science is taught through inquiry-based learning approaches with emphasis on the 5Es: engagement, exploration, explanation, elaboration and evaluation.

GENERAL SUBJECT LEARNING OUTCOMES

By the end of Junior Secondary school, the learner should be able to:

1. Acquire sufficient scientific knowledge, skills, values and attitudes to make informed choices on career pathways at Senior School and for everyday use, further education and training.
2. Select, improvise and safely use basic scientific apparatus, materials and chemicals effectively in everyday life.
3. Explore, manipulate, manage and conserve the environment for learning and sustainable development.
4. Practise relevant hygiene, sanitation and nutrition skills to promote good health.
5. Apply the understanding of body systems with a view to promoting and maintaining good health.
6. Develop capacity for scientific inquiry and problem solving in different situations.
7. Appreciate and use scientific principles and knowledge in everyday life.
8. Apply acquired scientific skills and knowledge to construct appropriate scientific devices from available resources.

STRAND 1.0: MIXTURES, ELEMENTS AND COMPOUNDS

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Questions
1.0. Mixtures, Elements and Compounds	1.1 **Properties of matter in the different states** (6 Hours)	By the end of the sub strand the learner should be able to: a) describe properties of the different states of matter, b) demonstrate diffusion in liquids, c) distinguish between temporary and permanent changes in substances, d) outline applications of change of state of matter in day-to-day life, e) appreciate the applications of change of state of matter in day-to-day life.	The learner is guided to: • perform simple experiments on properties of the different states of matter (*volume, shape, density, compressibility and ability to flow*), • perform experiments to demonstrate diffusion in liquids (*use of water and potassium manganate (VII*), • carry out simple experiments to demonstrate physical changes, temporary chemical changes and permanent changes of substances, • discuss the applications of change of state of matter in day-to-day life (*refrigerators, ice-cream vendors, fog formation, among others*), • where necessary, use digital devices to search, play and observe videos and animations showing the properties of different states of matter (in relation to *volume, shape, density, compressibility and ability to flow*).	How do the movement of particles in matter affect its physical properties?

Core competencies to be developed • Critical thinking and problem solving: developed as the learner discuss with peers the properties of the different states of matter. • Learning to learn: developed as the learner gains knowledge by manipulating apparatus and materials while carrying out simple experiments. • Digital literacy: as the learner searches, plays and watches videos on properties of different states of matter.
Pertinent and Contemporary Issues (PCIs) • Life skills: as they apply the knowledge on the change of state of matter in day-to-day life *(example, preparation of ice cubes, make candles*). • Financial literacy: as learners develop the economic awareness of the applications of change of state of matter in the locality.
Values: • Unity: as the learner carry out simple experiments in groups. • Respect: as the learner respects each other's opinion when discussing the applications of change of state of matter in day to day. • Responsibility: as the learner handles experimental equipment while performing simple experiments on properties of the different states of matter
Links to other subjects: • Home Science: in the preservation of foods by applying the knowledge of change of state of matter. • Physical Education and Sports: when ice cubes are used to relieve muscle pain in case of sprain.

Assessment Rubric				
Indicator	**Exceeds expectation**	**Meets expectation**	**Approaches expectation**	**Below expectation**
Ability to describe the properties of the different states of matter.	Correctly and consistently describes the properties of the different states of matter.	Correctly describes the properties of the different states of matter.	Correctly describes some of the properties of the different states of matter.	With assistance, describes properties of the different states of matter.
Ability to demonstrate diffusion in liquids.	Correctly and consistently demonstrates diffusion in liquids with illustrations.	Correctly demonstrates diffusion in liquids.	Partly demonstrates diffusion in liquids	With prompts, demonstrates diffusion in liquids.
Ability to distinguish between temporary and permanent changes of substances.	Correctly and consistently distinguishes between temporary and permanent changes in substances.	Correctly distinguishes between temporary and permanent changes in substances	Correctly distinguishes between some temporary and permanent changes in substances.	With assistance, distinguishes between temporary and permanent changes in substances.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Questions
1.0 Mixtures, Elements and Compounds	1.2 **Elements and Compounds** (7 Hours)	By the end of the sub strand the learner should be able to: a) distinguish between an element and a compound, b) relate common elements to their symbols, c) outline the applications of common elements in day-to-day life, d) appreciate the information on packaging labels of commonly consumed substances.	The learner is guided to: ● discuss the difference between elements and compounds, ● assign appropriate symbols to common elements and compounds cover (*copper*, *aluminium*, *iron,silver*, *table salt*, *and water*), ● discuss the names of common elements and their symbols (the first 13 elements of the periodic table and commonly used metals: *zinc*, *lead*, *tin*, *gold*, *mercury* and limited to the latin names only where applicable), ● discuss the importance and market value of common elements and compounds in society (*jewellery*, *iron*, *toiletries*, *food nutrients,mineral elements*, *medals among others*), ● Sample labelled containers of different substances indicating the common elements as part of the ingredients.	1. How are symbols assigned to elements? 2. What is the value of elements in day-to-day life?

Core competencies to be developed

- Learning to learn: developed as the learner identifies labels on containers indicating the common elements as part of the ingredients.

- Communication and collaboration: developed as the learner assigns symbols to common elements and compounds.

Pertinent and Contemporary Issues (PCIs)

- Financial literacy: as the learner discusses the importance and market value of common elements and compounds in society.

Values

- Respect and love: as the learner harmoniously work in a group to assign symbols to common elements and compounds.
- Responsibility: as the learner samples containers with labels while taking care of the environment.

Links to other subjects

- Home Science: when using ingredients and items made from the common elements and compounds.
- Business Studies: as the learner studies the market value of common elements and compounds.
- Agriculture: as the learner studies the mineral elements required by plants.

Assessment Rubric

Indicator	Exceeds expectation	Meets expectation	Approaches expectation	Below expectation
Ability to distinguish between an element and a compound.	Correctly and consistently distinguishes between an element and a compound.	Correctly distinguishes between an element and a compound.	Correctly distinguishes between some elements and compounds with assistance.	Requires assistance to distinguish between an element and a compound.
Ability to relate common elements to their symbols.	Correctly and consistently relates common elements to their symbols.	Correctly relates common elements to their symbols.	Correctly relates some of the common elements to their symbols with assistance.	With prompts, relates common elements to their symbols.
Ability to outline applications of common elements in society.	Correctly and consistently explains the applications of common elements.	Correctly outlines the applications of common elements.	Correctly outlines some of the applications of common elements.	With assistance, outlines applications of common elements.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Questions
1.0 **Mixtures, Elements and Compounds**	1.3 **Structure of the atom** (7 Hours)	By the end of the sub strand the learner should be able to: a) describe the structure of an atom and electron arrangement of elements, b) determine atomic number and mass number of elements, c) classify elements into metals and non-metals, d) appreciate the value of different elements in day-to-day life.	The learner is guided to: • discuss the meaning of the atom and illustrate its structure (protons, neutrons, and electrons), • draw and discuss the electron arrangements of elements and classify them into metals and non-metals (*first 20 elements of the periodic table*), • discuss and illustrate the atomic number and mass number of elements (*first 13 elements of the periodic table*), • use digital or print media to search for information on the structure of an atom, electron arrangement, atomic number and mass number of elements, • **Project:** model the atomic structure of selected elements of the periodic table using locally available materials.	1. What is the structure of an atom? 2. How do atoms gain stability?
Core competencies to be developed • Communication and collaboration: as the learner discusses with peers the meaning of atom. • Imagination and Creativity: as the learner uses locally available materials to model the atomic structure of selected elements in the periodic table.				

Pertinent and Contemporary Issues (PCIs

- Life skills: as the learner uses locally available materials to model the atomic structure of selected elements in the periodic table.

Values

- Respect and love: as the learner work with others in groups to illustrate the atomic number and mass number of elements.
- Responsibility: as the learner models the atomic structure of selected elements of the periodic table using locally available materials.
- Integrity: as the learner uses digital devices to search for information on the structure of an atom, electron arrangement, atomic number and mass number of elements.

Links to other subjects

- Mathematics: as they distribute electrons in the various energy levels of atoms of elements.

Assessment Rubric

Indicator	Exceeds expectation	Meets expectation	Approaches expectation	Below expectation
Ability to describe the structure of an atom and electron arrangement of selected elements.	Correctly and consistently describes the structure of an atom and electron arrangement of selected elements	Correctly describes the structure of an atom and electron arrangement of selected elements.	Correctly describes some of the structures of an atom and electron arrangement of some selected elements.	With prompts, describes the structure of an atom and electron arrangement of selected elements.
Ability to determine atomic number and mass number of elements.	Correctly and consistently determines atomic number and mass number of elements.	Correctly determines atomic number and mass number of elements.	Correctly determines the atomic number the mass number of some element.	With assistance, determines the atomic number and mass number of elements.

Ability to classify the selected elements into metals and non-metals.	Correctly and consistently classifies the selected elements into metals and non-metals.	Correctly classifies the selected elements into metals and non-metals.	Correctly classifies some of the selected elements into metals and non-metals.	Needs assistance to classify selected elements into metals and non-metals with assistance.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Questions
1.0 **Mixtures, Elements and Compounds**	1.4 **Oxygen** (7 Hours)	By the end of the sub strand the learner should be able to: a) prepare oxygen in the laboratory, b) investigate the physical and chemical properties of oxygen, c) explain the role of oxygen in combustion and spread of fire, d) identify classes of fire and their control measures, e) appreciate the role of oxygen in day to day life	The learner is guided to: • carry out experiment using hydrogen peroxide/potassium permanganate to prepare oxygen, • discuss the role of oxygen in combustion and the spread of fire, • classify fire according to the cause and suggest control measures, • practise fire control measures (*breaking the fire triangle and use of fire extinguishers*), • discuss rights to safety and access to information on flammable substances, • discuss the role of oxygen in every life.	1. How is oxygen important in day to day life? 2. What are the different classes of fire?

			• where possible, use digital devices to search, play and watch and discuss videos and animations on the different classes of fire.	

Core competencies to be developed

- Citizenship: as the learner practices fire control measures.
- Communication and collaboration: as the learner discusses with peers the rights to safety and access to information on flammable substances.
- Learning to learn: as the learner discusses with peers the role of oxygen in combustion and spread of fire.

Pertinent and Contemporary Issues (PCIs

- Disaster and risk reduction: as the learner applies different methods of fire control.

Values

- Respect and love: as the learner works with peers in groups to classify fire according to the cause and suggest control measures.
- Responsibility: as the learner discusses with peers the rights to safety and access to information on flammable substances

Links to other subjects

- Home Science: when they practice safety measures to prevent fire accidents.

Assessment Rubric

Indicator	**Exceeds expectation**	**Meets expectation**	**Approaches expectation**	**Below expectation**
Ability to prepare oxygen in the laboratory.	Systematically and correctly prepares oxygen in the laboratory.	Systematically prepares oxygen in the laboratory.	Prepares oxygen in the laboratory, but omits some steps.	With assistance, prepares oxygen in the laboratory.
Ability to	Correctly investigates	Correctly investigates	Investigates physical	With assistance,

investigate physical and chemical properties of oxygen.	physical and chemical properties of oxygen, giving examples.	physical and chemical properties of oxygen.	and chemical properties of oxygen, but omits some steps.	investigates physical and chemical properties of oxygen.
Ability to explain the role of oxygen in combustion and spread of fire.	Correctly and consistently explains the role of oxygen in combustion and spread of fire.	Correctly explains the role of oxygen in combustion and spread of fire.	Partially explains the role of oxygen in combustion and spread of fire.	Needs assistance to explain the role of oxygen in combustion and spread of fire
Ability to identify classes of fire and their control measures.	Correctly and consistently identifies classes of fire and their control measures.	Correctly identifies classes of fire and their control measures.	Correctly identifies some of the classes of fires and their control measures	Needs guidance to classes of fires and their control measures.

STRAND 2.0: LIVING THINGS AND THEIR ENVIRONMENT

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Questions
2.0 **Living Things and their Environment**	2.1 **The Cell** (18 Hours)	By the end of the sub strand, the learner should be able to: a) identify the components of a cell as seen under the light microscope and state their functions, b) compare plant and animal cells as observed under a light microscope, c) calculate the magnification of cells seen under the light microscope, d) appreciate the use of a light microscope in magnification.	The learner is guided to: • observe, identify, draw, label and state the functions of parts of the light microscope, • practise how to use and care for a light microscope, • prepare, mount and observe plant cells under a light microscope, • observe permanent slides of animal cells under the light microscope, • discuss with others, the differences between plant and animal cells as seen under a light microscope, • calculate magnification at various objective lenses of the light microscope.	1. Why is the light microscope important in day-to-day life? 2. What are the differences between plant and animal cells?
Core competencies to be developed: • Citizenship: as the learner works in harmony with peers in groups that enhance social cohesion. • Creativity and imagination: as the learner applies knowledge and skills gained to draw and label parts of the light microscope.				

- Self-efficacy: as the learner prepares and observes specimens under the light microscope.

Pertinent and Contemporary Issues (PCIs)

- Social cohesion: as the learner works together in groups to prepare, mount and observe plant cells under a light microscope.
- Environmental conservation: as the learner safely uses and disposes of used specimens and equipment.

Values

- Respect: as the learner respects each others' opinion when discussing the differences between plant and animal cells as seen under a light microscope.
- Responsibility: as the learners play different roles as they prepare, mount and observe plant cells under a light microscope.

Links to other subjects

- Agriculture: as learners learn about the role of diffusion and osmosis in plant nutrition.
- Mathematics: as the learner calculates magnification of cells as seen under a light microscope.

Assessment Rubric

Indicator	Exceeds expectation	Meets expectation	Approaches expectation	Below expectation
Ability to identify the components of a cell as seen under the light microscope and state their functions.	Correctly and consistently identifies all the components of a cell as seen under the light microscope and states their functions.	Correctly identifies all the components of a cell as seen under the light microscope and states their functions.	Correctly identifies some components of a cell as seen under the light microscope and states their functions.	Has difficulty identifying the components of a cell, as seen under the light microscope and stating their functions.

Ability to compare plant and animal cells as observed under a light microscope.	Correctly and consistently compares plant and animal cells as observed under a light microscope.	Correctly compares plant and animal cells as observed under a light microscope.	Correctly makes some comparisons between plant and animal cells with guidance.	Has difficulty comparing plant and animal cells as observed under a light microscope.
Ability to calculate the magnification of cells as seen under the light microscope	Correctly calculates the magnification of cells as seen under the light microscope with precision.	Correctly calculates the magnification of cells as seen under the light microscope	Calculates the magnification of cells as seen under the light microscope with some inaccuracies.	Has difficulty calculating the magnification of cells as seen under the light microscope.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Questions
2.0 **Living Things and their Environment**	2.2 **Movement of materials in and out of the cell** (32 Hours)	By the end of the sub strand the learner should be able to; a) describe the structure and properties of the cell membrane, b) explain the role of diffusion in living things c) demonstrate the process of osmosis in living things,	The learner is guided to; ● carry out activities to illustrate the structure of the cell membrane, ● brainstorm and discuss the properties of the cell membrane, ● carry out experiments to demonstrate the effects of heat, dilutes acids and alkalis on the cell membrane, ● carry out experiments to	1. How is diffusion and osmosis important in living organisms? 2. What are the similarities and differences

		d) describe factors affecting diffusion and osmosis, e) explain the roles of osmosis in living things, f) appreciate the importance of diffusion and osmosis in living things.	demonstrate diffusion using perfumes/scented flowers and discuss their roles in living things, ● carry out experiments on osmosis using plant materials and visking tubing, ● discuss with peers, the roles of osmosis in living things, ● observe and account for the changes that occur in the plant leaves at different times of the day, ● use digital or print media to search for information on the structure and properties of the cell membrane or watch videos and animations over the same, ● use digital or print media to search for information on factors that affect diffusion and osmosis, ● use digital or print media to search for information on how gases are exchanged in the human lungs.	between osmosis and diffusion?

Core competencies to be developed:

- **Communication and collaboration**: as the learner works with peers in groups to conduct experiments, prepare reports, present their findings using appropriate scientific language.
- **Citizenship**: as the learner works with peers in groups to demonstrate the effects of heat dilutes acids and alkalis on the cell

membrane that enhances social cohesion. • **Digital literacy**: as the learner uses various digital devices to search for information on the structure and properties of the cell membrane.
Pertinent and Contemporary Issues (PCIs) • **Social cohesion:** as the learner works with peers in groups to carry out experiments on osmosis using plant materials and visking tubing.
Values: • **Respect –** as the learner respects each other's ideas and opinion when discussin with peers in groups the roles of osmosis in living things. • **Responsibility-** as the learners play different role when carying out experiments on osmosis using plant materials and visking tubing.
Links to other subjects: • Agriculture: as learners learn about the role of diffusion and osmosis in plants and animals. • Computer Studies: as learners uses digital devices.

Assessment Rubric

Indicator	**Exceeds expectation**	**Meets expectation**	**Approaches expectation**	**Below expectation**
Ability to describe the structure and properties of the cell membrane.	Correctly and consistently describes the structure and properties of the cell membrane.	Correctly describes the structure and properties of the cell membrane.	Partly describes the structure and properties of the cell membrane.	Has difficulty describing the structure and properties of the cell membrane.

Ability to explain the role of diffusion in living things.	Correctly and comprehensively explains the role of diffusion in living things.	Correctly explains the role of diffusion in living things.	Partly explains the role of diffusion in living things.	Has difficulty explaining the role of diffusion in living things.
Ability to demonstrate the process of osmosis in living things.	Correctly demonstrates the process of osmosis in living things in a variety of ways.	Correctly demonstrates the process of osmosis in living things.	Partially demonstrates the process of osmosis in living things.	Has difficulty demonstrating the process of osmosis in living things.
Ability to describe factors affecting diffusion and osmosis.	Correctly describes factors affecting diffusion and osmosis, giving examples	Correctly describes factors affecting diffusion and osmosis.	Correctly describes some factors affecting diffusion and osmosis.	Has difficulty describing factors affecting diffusion and osmosis.
Ability to explain the roles of osmosis in living things.	Correctly explains the roles of osmosis in living things, giving real life examples.	Correctly explains the roles of osmosis in living things.	Partly explains the roles of osmosis in living things.	Has difficulty explaining the roles of osmosis in living things.

STRAND 3.0: FORCE AND ENERGY

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Questions
3.0 Force and Energy	**3.1 Transformation of energy** (9 Hours)	By the end of the sub strand, the learner should be able to; a) identify forms of energy in nature, b) explain energy transformations in nature, c) identify appliances whose working relies on energy transformation, d) describe safety measures associated with energy transformation, e) appreciate the applications of energy transformation in day-to-day life.	The learner is guided to: • discuss with peers and identify forms of energy found in the environment, • demonstrate and explain types of energy transformations using locally available materials, • discuss the processes energy transformation in day-to-day life (*electrical to heat, chemical to electrical, mechanical to electrical, electrical to light, electrical to sound and potential to kinetic*), • discuss applications of energy transformation in day-to-day life (*bulb, diodes, moving microphone, electric heater, solar panel, dynamo, motor*), • use digital or print media to	1. How can energy be transformed from one form to another? 2. How can energy transformation be applied in day-to-day life? 3. What are the energy transformation processes that occur in our environment?

			search for information on applications of energy transformation processes in day-to-day life or watch videos and animations on the same, • discuss with peers the safety measures associated with energy transformation in the environment, • use digital or print media to search for information on dangers associated with energy transformation and strategies of mitigating them (*relate to road accidents*; *K.E to P.E; action and reaction, accidents caused by fire, electricity, health hazard from bright light, loud sound*) or watch videos and animations showing the same	

Core competencies to be developed:

- Digital Literacy: as the learner searches for information on dangers associated with energy transformation and strategies of mitigating them.
- Communication and collaboration: as the learner discusses with peers the safety measures associated with energy

transformation in the environment.
Pertinent and Contemporary Issues (PCIs) • Sustainable Development: as the learner identifies and explains the applications of energy transformation in day-to-day life. • Life Skills and Value Education: as the learner identifies the applications of energy transformation in day-to-day life.
Values: • Respect: as the learner respects each other's opinion when working in groups to discuss the energy transformation processes. • Responsibility: as learner demonstrates types of energy transformations using locally available materials.
Links to other subjects: • Home Science: as the learner learn about applications of energy transformation processes in day-to-day life.

Assessment Rubric

Indicator	**Exceeds expectation**	**Meets expectation**	**Approaches expectation**	**Below expectation**
Ability to identify forms of energy in nature.	Correctly identifies all forms of energy in nature, giving examples from the locality.	Correctly identifies all forms of energy in nature.	Correctly identifies some forms of energy in nature.	With help, identifies some forms of energy in nature.
Ability to explain energy transformation in nature.	Correctly explains energy transformation in nature, giving supporting illustrations.	Correctly explains energy transformation in nature.	Explains energy transformation in nature but ignore some key concepts.	Has difficulty explaining energy transformation in nature.
Ability to identify appliances whose working relies on energy transformation.	Correctly identifies appliances whose working relies on energy transformation, giving examples from the locality.	Correctly identifies appliances whose working relies on energy transformation.	Correctly identifies some appliances whose working relies on energy transformation	With prompt, identifying appliances whose working relies on energy transformation.

Ability to describe safety measures associated with energy transformation.	Correctly describes safety measures associated with energy transformation, giving examples.	Correctly describes safety measures associated with energy transformation.	Correctly describes some safety measures associated with energy transformation.	With prompt, describes safety measures associated with energy transformation.

Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Key Inquiry Questions**
3.0 Force and Energy	**3.2. Pressure I** (9 Hours)	By the end of the sub strand the learner should be able to; a) explain the meaning of pressure as used in science, b) describe pressure in solids and liquids, c) determine pressure in solids and liquids, d) identify applications of pressure in solids and liquids, e) Appreciate the applications of pressure	The learner is guided to: • brainstom and discuss with peers the meaning of pressure, • carry out activities to demonstrate and describe pressure in solids and liquids, • carry out experiments to determine pressure in solids and liquid (pressure exerted by objects with different surface areas e.g. bricks, pressure in liquids increases with depth and density), • solve numrical problems involving pressure (Pressure = Force/Area $\left(P = \frac{F}{A}\right)$ and Pressure = liquid density x gravitational	1. What is the meaning of pressure as used in science? 2. What are the applications of pressure in solids and liquids?

		in solids and liquids.	pull x column height of the liquid ($P = \rho g h$), • discuss and identify the applications of pressure in solids and liquids • use digital or print media to search for information on pressure in solids, liquids and their applications.	

Core competencies to be developed:

- Digital literacy as the learner searches for information on pressure in solids, liquids and their applications.
- Communication and collaboration as the learner discusses with peers the meaning of pressure as used in science**.**

Pertinent and Contemporary Issues (PCIs)

- Social cohesion: as the learner works with peers in groups to carry out experiments to determine pressure in solids and liquid.

Values:

- Responsibility: as the learners play different roles as they carry out experiments to determine pressure in solids and liquid.
- Unity: as the learner brainstoms and discusses with peers in a group, the meaning of pressure.

Links to other subjects:

- Wood Technology: as the learner the learner relates pressure exerted by an object through its surface area.

Assessment Rubric

Indicator	**Exceeds Expectation**	**Meets Expectation**	**Approaches Expectation**	**Below Expectation**
Ability to explain the meaning of pressure as used in	Correctly and comprehensively explains the meaning of	Correctly explains the meaning of pressure as used in science.	Correctly explains the meaning of pressure as used in science.	Has challenges explaining the meaning of pressure as

science.	pressure as used in science.			used in science.
Ability to describe pressure in solids and liquids.	Correctly describes the pressure in solids and liquids, giving relevant illustrations.	Correctly describes pressure in solids and liquids.	Describes pressure in solids and liquids, leaving out some key points.	With prompt, describes pressure in solids and liquids.
Ability to determine pressure in solids and liquids.	Correctly determines pressure in solids and liquids with ilustrations.	Correctly determines pressure in solids and liquids.	Correctly determines either pressure in solids or in liquids.	With prompt, determines pressure in solids and liquids.
Ability to identify applications of pressure in solids and liquids.	Correctly and consistently identifies applications of pressure in solids and liquids.	Correctly identifies applications of pressure in solids and liquids.	Identifies some applications of pressure in solids and liquids	With help, identifies applications of pressure in solids and liquids.

COMMUNITY SERVICE-LEARNING PROJECT

Introduction

In Grade 8, focus is on learners making preparations to undertake a CSL activity of their own choice. They will be required to identify a community problem through research, plan and come up with solutions to solve the problem. The preparations will be carried out in groups. Learners will build on CSL knowledge, skills and attitudes acquired from Life Skills Education as well as other subjects.

CSL Skills to be Covered:

i) **Leadership**: Learners develop leadership skills as they undertake various roles during preparation.

ii) **Financial Literacy and Entrepreneurship Skills:** Learners will gain skills on wise spending, saving and investing for sustained economic growth. They could consider ways of generating income as they undertake the CSL project through innovative ways. Moreover, they could identify business ideas and opportunities as well as resources to meet the needs of the community.

iii) **Research:** Learners will be expected to identify a problem or pertinent issue in the community and indicate how the problem will be solved. They will also acquire skills on how to report their findings.

iv) **Communication:** Learners indicate reporting mechanisms to be used during the actual project e.g., how they intend to communicate with members of the community, either online or offline.

v) **Citizenship:** As learners engage in the CSL activities for this Grade, they will be vested with the rights, privileges and duties of a citizen, hence giving them a sense of belonging and attachment to the nation. They will also be empowered to engage and assume active roles in shaping a more peaceful, tolerant and inclusive society.

vi) **Life Skills Education:** Learners will be equipped with life skills, including decision making, assertiveness, effective communication, problem solving and stress management. This will enable them to manage interpersonal relationships, develop leadership skills as well as discover and grow their talents.

vii) **Community Development:** Learners will be empowered with skills necessary to effect relevant change, including building stronger and more resilient communities.

Suggested PCIs	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Questions
• Environmental degradation • Lifestyle diseases • Communicable and non-communicable diseases • Poverty • Violence in community • Food security issues • Conflicts in the community **Note:** The suggested PCIs are only examples. Teachers should allow learners to identify PCIs as per their context and reality.	By the end of the CSL project, the learner should be able to: a) identify a problem in the community through research b) plan to solve the identified problem in the community, c) design solutions to the identified problem, d) appreciate the need to belong to a community.	The learner is guided to: • brainstorm on pertinent and contemporary issues in their community that need attention in groups • choose a PCI that needs immediate attention and explain why in groups • carry out research using digital devices/print media/interactions with members of the community/resource persons in identifying a community problem to address in groups • discuss possible solutions to the identified issue in groups • propose the most appropriate solution to the problem in groups • discuss ways and instruments they can use to collect data on the problem (*questionnaires, interviews, observation schedule, etc*) • develop instruments for data collection • identify resources needed for the CSL project (*human, technical, financial*) • discuss when the project will begin and end	1. How does one determine community needs? 2. Why is it necessary to make adequate preparations before embarking on a project?

		• prepare a programme/timetable of the entire project execution • Assign roles to be carried by all group members • reflect on how the project preparation enhanced learning.	

Key Component of CSL developed

a) Identification of a problem in the community through research
b) planning to solve the identified problem
c) designing solutions to the identified problem

Core competencies to be developed

- Communication and collaboration: Learners will make the preparations in groups and conduct discussions on best ways of carrying out the project.
- Self-efficacy: Learners develop the skills of self-awareness and leadership as they undertake the CSL project
- Creativity and imagination: Learners will come up with creative ways of solving the identified community problem
- Critical thinking and problem solving: Learners will demonstrate autonomy in identifying a community need, exploring plausible solutions and making necessary preparations to address the problem.
- Digital literacy: Learners can use technology as they research on a community problem that they can address.
- Learning to learn: Learners gain new knowledge and skills as they identify a community problem to be addressed and make preparations to carry out the project.
- Citizenship: This is enhanced as learners choose a PCI that needs immediate attention in the community.

Pertinent and contemporary Issues

- Social cohesion as learners discuss possible solutions to the identified issue.
- Critical thinking as learners discuss possible solutions to the identified issue.

Values

- Integrity as learners carry out research using digital devices and print media as they identify a community problem to address.

- Respect as learners brainstorm on pertinent and contemporary issues in their community that need attention

Assessment Rubric

Indicator	Exceeds Expectation	Meets Expectation	Approaches Expectation	Below Expectation
Ability to identify a problem in the community through research	Correctly and precisely identifies a problem in the community through research	Correctly identifies a problem in the community through research	Partially identifies a problem in the community through research	Partially identifies a problem in the community through research with assistance
Ability to plan to solve the identified problem	Accurately and systematically plans to solve the identified problem	Accurately plans to solve the identified problem	Plans to solve the identified problem leaves out some details	With assistance plans to solve the identified problem but leaves out many details
Ability to design solutions to the identified problem	Correctly and elaborately designs solutions to the identified problem	Correctly designs solutions to the identified problem	Partly designs solutions to the identified problem	Partly designs solutions to the identified problem with prompting

APPENDIX: LIST OF ASSESSMENT METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES

Assessment Methods in Science	Learning Resources	Non-Formal Activities
• Reflections • Game Playing • Pre-Post Testing • Model Making • Explorations • Experiments • Investigations • Conventions, Conferences, and Debates • Applications • Teacher Observations • Project • Journals • Portfolio • Oral or Aural Questions • Learner's Profile • Written Tests • Anecdotal Records	• Laboratory Apparatus and Equipment • Textbooks • Software • Relevant reading materials • Digital Devices • Recordings	• Visit the science historical sites. • Use digital devices to conduct scientific research. • Organizing walks to have live learning experiences. • Developing simple guidelines on how to identify and solve some community problems. • Conducting science document analysis. • Participating in talks by resource persons on science concepts. • Participating in science clubs and societies • Attending and participating science and engineering fairs • Organizing and participating in exchange programmes. • Making oral presentations and demonstrations on science issues.